# PROJECT PROPOSAL

**Master of Information Technology — Independent Study (MIT4301)**

## 1. Project Title

Automated Shared Vehicle Reservation and Fleet Management System

## 2. Background and Problem Understanding

In many organizations, managing shared vehicles, assigning drivers, and tracking vehicle maintenance are handled manually using logbooks or isolated spreadsheets. This traditional approach leads to several operational challenges:

- **Double Booking:** Multiple employees accidentally booking the same vehicle for overlapping times.
- **Maintenance Oversight:** Vehicles missing routine service intervals due to lack of an automated alert system.
- **Inaccurate Trip Logs:** Missing or delayed tracking of mileage (start/end KM), making fuel and usage optimization difficult.

## 3. Project Objectives

The primary objectives of the proposed system are:

1. To develop a user-friendly web interface allowing employees to check availability and book shared vehicles efficiently.
2. To provide administrative tools for managers to easily allocate drivers and approve booking requests.
3. To automate trip log calculations based on initial and final mileage input.
4. To restrict vehicle booking capabilities during designated maintenance schedules to prevent breakdown risk.

## 4. Proposed System

The system will operate with three distinct user roles: Administrators, Employees, and Drivers. Employees can request reservations; Administrators manage vehicle inventory, assign drivers, and handle approvals; and Drivers log essential trip information like mileage upon journey completion.

## 5. Project Scope

**In-Scope:**

- User registration, role-based login, and profile management.
- Vehicle availability checks, conflict prevention, and dynamic booking forms.
- Driver allocation module and status-based journey tracking (Pending, Approved, Ongoing, Completed).
- Trip logging system capturing odometer readings and generating auto-calculated mileage summaries.
- Vehicle maintenance blockouts preventing bookings during service intervals.

**Out-of-Scope:**

- Live real-time GPS tracking on map widgets (to maintain a feasible sifting threshold within the 10-week cycle).
- Automated payment gateway integration or live fuel price monitoring APIs.

## 6. Proposed Technologies and Tools

- **Frontend:** HTML5, CSS3, JavaScript, Bootstrap framework (for clean UI/UX rendering).
- **Backend & Server:** PHP (object-oriented script handling) and Apache server environment.
- **Database Store:** MySQL (Relational management store local layout).
- **Version Control:** Git and GitHub for structural repository mapping.

## 7. Proposed Development Approach

The system implementation will follow a structured process: Requirements analysis will build functional flow diagrams, design phases will layout the MySQL tables, development will implement features incrementally week-over-week, followed by local system validation testing before delivery.

## 8. Expected Deliverables

- Fully functional database-backed information system source code.
- Comprehensive weekly structural development portfolio (Chapters 1 to 10).

## 9. Work Plan / Timeline

| Weeks | Milestones / Deliverables |
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| Weeks 1-2 | Topic analysis, boundary setup, and proposal sign-off. |
| Weeks 3-5 | UI layouts, database normalization mapping, tool setup. |

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| Weeks 6-8 | Feature scripting, core allocation triggers, trip logging functions. |
| Weeks 9-10 | System boundary error testing, clean-up, refactoring code. |
| Weeks 11-13 | Assembly of the final 10-chapter portfolio, Viva prep. |
| Weeks 14-15 | Live system presentation and oral defense panel review. |

# 10. References


1. Oracle Corporation, 2026. *MySQL Reference Manual*. Available online.
2. The PHP Group, 2026. *PHP Documentation Manual*. Available online.